

Part B

Q4 (a)

To prove that the family $L(X, \theta)$ has a monotone likelihood ratio (MLR) in $T = \sum_{i=1}^n (\log X_i)^2$ we need to show that for any $\theta_1 < \theta_2$ the likelihood ratio $\frac{L(X, \theta_2)}{L(X, \theta_1)}$ is a non decreasing function of T .

Step 1 Write down the likelihood function $L(X, \theta)$
Random Variables $X_1, \dots, X_n$ are i.i.d. with density function

$$f(x, \theta) = \frac{1}{x\theta\sqrt{2\pi}} \exp\left(-\frac{1}{2} \left[\frac{\log(x)}{\theta}\right]^2\right)$$

The likelihood function for sample $X = (X_1, \dots, X_n)$ is the product of the individual densities.

$$\begin{aligned} L(X, \theta) &= \prod_{i=1}^n f(x_i, \theta) = \prod_{i=1}^n \left(\frac{1}{x_i \theta \sqrt{2\pi}} \exp\left(-\frac{1}{2} \left[\frac{\log(x_i)}{\theta}\right]^2\right) \right) \\ &= \left(\frac{1}{\theta \sqrt{2\pi}} \right)^n \left(\prod_{i=1}^n \frac{1}{x_i} \right) \exp\left(-\frac{1}{2\theta^2} \sum_{i=1}^n (\log x_i)^2\right) \end{aligned}$$

$$\text{Let } T = \sum_{i=1}^n (\log x_i)^2$$

$$\therefore L(X, \theta) = C(X) \left(\frac{1}{\theta \sqrt{2\pi}} \right)^n \exp\left(-\frac{T}{2\theta^2}\right) \quad \text{where } C(X) = \prod_{i=1}^n \frac{1}{x_i}$$

$C(X)$ independent of θ .

Step 2 From the likelihood ratio:

Let $\theta_1 < \theta_2$ $\therefore$ likelihood ratio

$$\frac{L(X, \theta_2)}{L(X, \theta_1)} = \frac{C(X) \left(\frac{1}{\theta_2 \sqrt{2\pi}} \right)^n \exp\left(-\frac{T}{2\theta_2^2}\right)}{C(X) \left(\frac{1}{\theta_1 \sqrt{2\pi}} \right)^n \exp\left(-\frac{T}{2\theta_1^2}\right)} = \left(\frac{\theta_1}{\theta_2} \right)^n \exp\left(T \left(\frac{1}{2\theta_1^2} - \frac{1}{2\theta_2^2} \right)\right)$$

Step 3 Analyze Monotonicity

Let $g(T) = \frac{L(X, \theta_2)}{L(X, \theta_1)}$. We need to show $g(T)$ is a non decreasing function of T

Consider coefficient of T in the exponent.

$$K = \frac{1}{2\theta_1^2} - \frac{1}{2\theta_2^2}$$

Since $\theta_1 < \theta_2 \therefore \theta_1^2 < \theta_2^2$
 $\therefore \frac{1}{\theta_2^2} > \frac{1}{\theta_1^2} \therefore K = \frac{1}{2\theta_1^2} - \frac{1}{2\theta_2^2} > 0$

Part B

Q4 (a)

Contd:

$$g(T) = \left(\frac{\theta_1}{\theta_2}\right)^n e^{KT}$$

θ_1/θ_2 is a +ve constant.

$\therefore K > 0$ e^{KT} is an increasing function of T

$\therefore g(T)$ is a non-decreasing function of T .

Conclusion:

Since Likelihood ratio $\frac{L(X, \theta_2)}{L(X, \theta_1)}$ is a non-decreasing function of $T = \sum_{i=1}^n (\log X_i)$ for any $\theta_1 < \theta_2$, the

family $\{L(X, \theta); \theta > 0\}$ has a monotone likelihood ratio in T . Hence the statement is proved.

Part B
84 (b)

Argument for the existence of a UMP test:

The existence of Uniformly Most Powerful (UMP) test for one-sided hypothesis (like $H_0: \theta \leq \theta_0$ vs $H_1: \theta > \theta_0$) is guaranteed by Karlin-Rubin theorem (Monotone likelihood theorem)

It says, if a family of probability distributions has a monotone likelihood ratio (MLR) in a statistic $T(X)$, then for testing $H_0: \theta \leq \theta_0$ vs $H_1: \theta > \theta_0$ there exists a UMP test of size α

$\therefore$ from part (a) the likelihood ratio is increasing in T .
 $\therefore$ a UMP α -size test exists for $H_0: \theta \leq \theta_0$ vs $H_1: \theta > \theta_0$

Structure of the UMP test:

The UMP test reject H_0 if the MLR statistic $T(X)$ is sufficiently large.

$\therefore$ structure of UMP α -size test $\varphi(x)$ is

$$\varphi^*(x) = \begin{cases} 1 & \text{if } T > c \\ \gamma & \text{" } T = c \\ 0 & \text{" } T < c \end{cases}$$

where: $T = \sum_{i=1}^n (\log \frac{f(x_i)}{f_0(x_i)})^2$ is the test statistic

$\rightarrow$ 'c' is a threshold constant (critical value) determined by the size α of the test, such that

$$P_{\theta_0}(T > c) + \gamma P_{\theta_0}(T = c) = \alpha$$

$\rightarrow \gamma$ is a randomization probability. (Typically = 0 for continuous distributions)

For continuous distribution, $P(T = c) = 0$, the test simplifies to:

$$\varphi(x) = \begin{cases} 1 & \text{if } \sum_{i=1}^n (\log X_i)^2 > c \\ 0 & \text{if } \sum_{i=1}^n (\log X_i)^2 \leq c \end{cases}$$

~~Not a valid test for null hypothesis H_0~~

Part B

Q4 (b)

Contd:

$\therefore$ This test rejects the null Hypothesis H_0 if the observed value of the statistic $T = \sum_{i=1}^n (\log X_i)^2$ is greater than some critical value 'c'.

So, Yes, there is a UMP α -size test for $H_0: \theta \leq \theta_0$ vs $H_1: \theta > \theta_0$; it rejects H_0 when $T = \sum_{i=1}^n (\log X_i)^2$ is sufficiently large.

Part B
Q4 c)

Given density of X_i :

$$f_X(x, \theta) = \frac{1}{x\theta\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left[\frac{\log(x)}{\theta}\right]^2\right), x > 0.$$

we want to find distribution of $Y_i = \log X_i$

Let $Y = \log X$ $\therefore$ the inverse is $X = e^Y$

$$\therefore \left|\frac{dx}{dy}\right| = \left|\frac{d(e^Y)}{dY}\right| = |e^Y| = e^Y \quad \left|\frac{dx}{dy}\right| = e^Y$$

$$\therefore \text{density for } f_Y(y) = f_X(e^Y, \theta) \cdot e^Y$$

$$f_Y(y) = \frac{1}{e^Y \theta \sqrt{2\pi}} \exp\left(-\frac{1}{2}\left[\frac{\log(e^Y)}{\theta}\right]^2\right) \cdot e^Y$$

$$= \frac{1}{\theta \sqrt{2\pi}} \exp\left(-\frac{Y^2}{2\theta^2}\right), -\infty < Y < \infty$$

$\Rightarrow$ Normal Probability density function.

$$f_{\text{Normal}}(y; \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right)$$

$$\therefore \text{ if } \mu = 0 \text{ \& } \sigma^2 = \theta^2$$

$$f_Y(y) = f_{\text{Normal}}(y; \mu, \sigma^2)$$

$\therefore Y_i = \log X_i$ Has a Normal distribution with mean = 0 & variance = θ^2 ~~and log~~

$$\therefore Y_i = \log X_i \sim N(0, \theta^2)$$

Hence proved

Part B
Q4 (d)

UMP α -size test φ^* rejects Null Hypothesis $H_0: \theta \leq \theta_0$
when test statistic $T > C$

$$\varphi^*(x) = \begin{cases} 1 & \text{if } T > C \\ 0 & \text{if } T \leq C \end{cases}$$

When $\theta = \theta_0$ $Y_i = \log X_i \sim N(0, \theta_0^2)$

$$Z_i = Y_i / \theta_0 = \frac{\log X_i}{\theta_0} \sim N(0, 1)$$

$$Z_i^2 = \left(\frac{\log X_i}{\theta_0} \right)^2 \sim \chi_1^2 \rightarrow \text{Chi-sqr distr. with 1 degree of freedom}$$

Since X_i are i.i.d., Z_i^2 are also i.i.d. χ_1^2 variables

Test statistic T is sum of these squared terms, scaled by θ_0^2

$$T = \sum_{i=1}^n (\log X_i)^2 = \theta_0^2 \sum_{i=1}^n \left(\frac{\log X_i}{\theta_0} \right)^2 = \theta_0^2 \sum_{i=1}^n Z_i^2$$

Sum of n i.i.d. χ_1^2 variables is a χ_n^2 variable:

$$\sum_{i=1}^n Z_i^2 \sim \chi_n^2$$

$$T \sim \theta_0^2 \chi_n^2$$

$\therefore$ Under $H_0: \theta = \theta_0$:

$$\text{Size of the test is } \alpha \Rightarrow P_{\theta_0}(\text{Reject } H_0) = \alpha$$

$$\Rightarrow P_{\theta_0}(T > C) = \alpha$$

$$\Rightarrow P_{\theta_0}(\theta_0^2 \chi_n^2 > C) = \alpha$$

$$\Rightarrow P_{\theta_0}(\chi_n^2 > \frac{C}{\theta_0^2}) = \alpha$$

Let $\chi_{n,\alpha}^2$ be the upper α -quantile of the chi squared distribution with n degrees of freedom, i.e.

$$P(\chi_n^2 > \chi_{n,\alpha}^2) = \alpha \quad \therefore \text{to find the critical value } C$$

$$\frac{C}{\theta_0^2} = \chi_{n,\alpha}^2$$

$$\therefore C = \theta_0^2 \chi_{n,\alpha}^2$$

P.T.O.

Part B

Q4 (1)

Contd

Substituting threshold constant c into the test structure

$$\varphi^*(X) = \begin{cases} 1 & \text{if } \sum_{i=1}^n (\log X_i)^2 > \theta_0^2 X_{n,\alpha}^2 \\ 0 & \text{if } \sum_{i=1}^n (\log X_i)^2 \leq \theta_0^2 X_{n,\alpha}^2 \end{cases}$$

$$\therefore \varphi^*(X) = \begin{cases} 1 & \text{if } \frac{1}{\theta_0^2} \sum_{i=1}^n (\log X_i)^2 > X_{n,\alpha}^2 \\ 0 & \text{if } \frac{1}{\theta_0^2} \sum_{i=1}^n (\log X_i)^2 \leq X_{n,\alpha}^2 \end{cases}$$

This completely defines the uniformly most powerful α -size test for $H_0: \theta \leq \theta_0$ vs $H_1: \theta > \theta_0$.

Part B
Q4 (e)

Calculation of the Power Function:

Power(θ) or $E_{\theta}[\varphi^*(x)]$ is the probability of rejecting the null hypothesis H_0 when the parameter value is θ .

The UMP test φ^* rejects $H_0: \theta \leq \theta_0$ when the statistic $T = \sum_{i=1}^n (\log X_i)^2$ is greater than the critical value $c = \theta_0^2 X_{n,\alpha}^2$.

$\therefore$ The power function is: $\text{Power}(\theta) = P_{\theta}(\text{Reject } H_0) = P_{\theta}(T > c)$

Also we know that for any true value of θ , the statistic T/θ^2 follows a chi-squared distribution with n degrees of freedom.

$$\begin{aligned} \frac{T}{\theta^2} &\sim \chi_n^2 \\ \therefore \text{Power}(\theta) &= P_{\theta}\left(\frac{T}{\theta^2} > \frac{c}{\theta^2}\right) = P_{\theta}\left(\chi_n^2 > \frac{\theta_0^2 X_{n,\alpha}^2}{\theta^2}\right) \\ &= P_{\theta}\left(\chi_n^2 > \left(\frac{\theta_0}{\theta}\right)^2 X_{n,\alpha}^2\right) \end{aligned}$$

Let $F_{\chi_n^2}(\cdot)$ be the CDF of the chi-squared distribution with n degrees of freedom.

$$\therefore P(\chi_n^2 > x) = 1 - F_{\chi_n^2}(x)$$

$\therefore$ The power function can be written as:

$$\text{Power}(\theta) = 1 - F_{\chi_n^2}\left(\left(\frac{\theta_0}{\theta}\right)^2 X_{n,\alpha}^2\right)$$

Sketch Power function Graph:-

Part B

Q4 (e)

Contd:

~~Let $x = \theta_0$ Power(θ_0) = $1 - F_{X_n^2}(\frac{\theta_0^2}{\theta^2} X_{n,\alpha}^2)$ For $\theta = \theta_0$, $\frac{\theta_0^2}{\theta^2}$ is large $\rightarrow$ CDF near 1~~

* Domain:- Parameter $\theta > 0$

* Monotonicity:- Let $u(\theta) = \frac{\theta_0^2}{\theta^2} X_{n,\alpha}^2$. as θ increases $\uparrow u(\theta)$ decreases $\downarrow$
Since $F_{X_n^2}(x)$ is a non decreasing function of x , $1 - F_{X_n^2}(x)$ is a non-increasing function of x .

$\therefore$ Power(θ) = $1 - F_{X_n^2}(u(\theta))$ is strictly increasing

* Value at Null Hypothesis Boundary ($\theta = \theta_0$):- Power is the size of test α

$$\text{Power}(\theta_0) = 1 - F_{X_n^2}\left(\frac{\theta_0^2}{\theta_0^2} X_{n,\alpha}^2\right) = 1 - F_{X_n^2}(X_{n,\alpha}^2)$$

$$\text{By definition of } X_{n,\alpha}^2 \Rightarrow 1 - F_{X_n^2}(X_{n,\alpha}^2) = P(X_n^2 > X_{n,\alpha}^2) = \alpha$$

$\therefore$ the graph passes through the point (θ_0, α)

* As $\theta \rightarrow 0^+$ (Deep within H_0 or near boundary):-

$$\frac{\theta_0^2}{\theta^2} X_{n,\alpha}^2 \text{ approaches } +\infty$$

$$\therefore \lim_{\theta \rightarrow 0^+} \text{Power}(\theta) = 1 - F_{X_n^2}(+\infty) = 1 - 1 = 0.$$

Power approaches 0

* As $\theta \rightarrow +\infty$ (Deep within H_1):- $\frac{\theta_0^2}{\theta^2} X_{n,\alpha}^2$ approaches 0.

$$\therefore \lim_{\theta \rightarrow +\infty} \text{Power}(\theta) = 1 - F_{X_n^2}(0) = \frac{1}{n} P(X_n^2 \leq 0) = \frac{1}{n} \cdot 0 = 0 \quad \left[\because X_n^2 \text{ is a continuous distribution starting at } 0 \right]$$

$$\therefore \lim_{\theta \rightarrow +\infty} \text{Power}(\theta) = 1.$$

Power approaches 1

Part B

Q4 (e)

Contd 3:

* Power function starts at 0 for θ close to 0.

* Monotonically increases

* Crosses point (θ_0, α)

* Asymptotically approaches 1 as $\theta \rightarrow \infty$

$\therefore$ This is characteristically an S shaped curve for one-sided tests.

* The region corresponding to $\theta \leq \theta_0$ is the null hypothesis.

* " " " " $\theta > \theta_0$ " " Alternate " "

* The power should be at most α for $\theta \leq \theta_0$ (with equality at $\theta = \theta_0$)
and greater than α for $\theta > \theta_0$.

